

between two species in which one organism lives inside the cell or cells of another organism (the host). In particular, as we discussed in Concept 25.3, structural, biochemical, and DNA sequence data indicate that mitochondria and plastids are derived from bacteria that were engulfed by the ancestors of early eukaryotic cells. The evidence also suggests that mitochondria evolved before plastids. Thus, a defining moment in the origin of eukaryotes occurred when a host cell engulfed a bacterium that would later become an organelle found in all eukaryotes—the mitochondrion.

To determine which bacterial lineage gave rise to mitochondria, researchers have compared the DNA sequences of

mitochondrial genes (mtDNA) to those found in major clades of bacteria. In the **Scientific Skills Exercise**, you will interpret one such set of DNA sequence comparisons. Collectively, such studies indicate that mitochondria arose from an alpha proteobacterium (see Figure 27.17). Results from mtDNA sequence analyses also indicate that the mitochondria of protists, animals, fungi, and plants descended from a single common ancestor, thus suggesting that mitochondria arose only once over the course of evolution. Similar analyses provide evidence that plastids descended from a single common ancestor—a cyanobacterium that was engulfed by a eukaryotic host cell.

Scientific Skills Exercise

Interpreting Comparisons of Genetic Sequences

Which Bacteria Are Most Closely Related to Mitochondria? Early eukaryotes acquired mitochondria by endosymbiosis: A host cell engulfed an aerobic bacterium that persisted within the cytoplasm to the mutual benefit of both cells. In studying which living bacteria might be most closely related to mitochondria, researchers compared DNA sequences encoding ribosomal RNA (rRNA), a component of ribosomes. Ribosomes perform critical cell functions. Hence, genes encoding rRNA are under strong selection and change slowly over time, making them suitable for comparing even distantly related species. In this exercise, you will interpret some of the research data to draw conclusions about the phylogeny of mitochondria.

How the Research Was Done Researchers isolated and cloned DNA sequences from the gene that codes for the small-subunit rRNA molecule for the wheat (a eukaryote) mitochondrion and for five bacterial species:

- Wheat, used as the source of genes encoding mitochondrial rRNA
- *Agrobacterium tumefaciens*, an alpha proteobacterium that lives within plant tissue and produces tumors in the host
- *Comamonas testosteroni*, a beta proteobacterium
- *Escherichia coli*, a well-studied gamma proteobacterium that inhabits human intestines
- *Mycoplasma capricolum*, a gram-positive mycoplasma, which is the only group of bacteria lacking cell walls
- *Anacystis nidulans*, a cyanobacterium

Data from the Research Gene sequences encoding rRNA for the wheat mitochondrion and the five bacteria were aligned and compared. The data table below, called a *comparison matrix*,

	Wheat mitochondrion	<i>A. tumefaciens</i>	<i>C. testosteroni</i>	<i>E. coli</i>	<i>M. capricolum</i>	<i>A. nidulans</i>
Wheat mitochondrion	–	48	38	35	34	34
<i>A. tumefaciens</i>		–	55	57	52	53
<i>C. testosteroni</i>			–	61	52	52
<i>E. coli</i>				–	48	52
<i>M. capricolum</i>					–	50
<i>A. nidulans</i>						–

Data from D. Yang et al., Mitochondrial origins, *Proceedings of the National Academy of Sciences USA* 82:4443–4447 (1985).



► Wheat, used as the source of mitochondrial RNA

summarizes the comparison of 617 nucleotide positions from the gene sequences. Each value in the table is the percentage of the 617 nucleotide positions for which the pair of organisms have the same base. Any positions that were identical across all six RNA-encoding gene sequences were omitted from this comparison matrix.

INTERPRET THE DATA

1. First, make sure you understand how to read the comparison matrix. **(a)** Find the cell that represents the comparison of *C. testosteroni* and *E. coli*. What value is given in this cell? **(b)** What does that value signify about the comparable gene sequences in those two organisms? **(c)** Explain why some cells have a dash rather than a value. **(d)** Why are some cells shaded gray, with no value?
2. Why did the researchers choose one plant mitochondrion and five bacterial species to include in the comparison matrix?
3. **(a)** Which bacterium has an rRNA-encoding gene that is most similar to that of the wheat mitochondrion? **(b)** What is the significance of this similarity?

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.